Depth-Guided YOLO Framework for Robust Pothole Detection under Low-Light and Rainy Conditions

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DATE:2025

Abstract

Road safety remains a major concern in developing and urban regions where potholes pose serious threats to vehicles and commuters, especially during adverse weather conditions. Traditional computer vision systems often fail to maintain accuracy under low-light and rainy environments due to poor visibility, reflections, and occlusions. This study proposes a Depth-Guided YOLO Framework that integrates RGB imagery with depth data to enhance the robustness and precision of pothole detection. By fusing visual and depth cues, the system improves target localization and distinguishes true potholes from water puddles and shadows. The proposed model is evaluated on real-world datasets captured in rainy and low-light scenarios, showing significant improvement in detection accuracy and reliability compared to standard YOLO architectures. The framework demonstrates its potential for real-time deployment in smart vehicles and intelligent transportation systems (ITS) for safer navigation.

Keywords

Pothole detection; YOLO; depth-guided imaging; low-light vision; rainy conditions; real-time detection; computer vision; intelligent transportation systems.

1. Introduction

Potholes are one of the most prevalent causes of road surface deterioration, leading to vehicle damage, traffic congestion, and accidents. Detecting these anomalies in real-time is crucial for improving road maintenance efficiency and enhancing vehicular safety. While conventional deep learning methods like the You Only Look Once (YOLO) framework have achieved notable success in object detection, they struggle in challenging visual environments such as rain, fog, and low-light conditions. Low illumination and rainfall often introduce noise, reflections, and inconsistent textures, making 2D detection models prone to false positives and missed detections. To overcome these limitations, this research introduces a Depth-Guided YOLO Framework, which incorporates 3D spatial information to complement RGB data. The integration of depth sensing allows the model to better differentiate surface irregularities, such as potholes, from visual artifacts like shadows or wet surfaces. This approach ensures improved robustness and accuracy, paving the way for reliable pothole detection in real-world driving conditions.

2. Methodology

The proposed system consists of three main stages: data acquisition, feature extraction, and depth-guided detection.

Table 1. Overview of Methodological Steps

Stage	Description	Tools/Techniques Used
Data Acquisition	Capturing synchronized RGB and depth images under varied lighting and weather conditions.	RGB-D camera, Rain simulation dataset, low-light enhancement filters.
Preprocessing	Image denoising, normalization, and contrast enhancement for better feature visibility.	CLAHE, Gaussian smoothing, depth normalization.
Feature Extraction	Combining visual and geometric cues to detect potential road defects.	Depth-guided fusion of RGB channels with disparity maps.
Model Architecture	Modifying YOLOv8 backbone to accept dual input (RGB + Depth).	Depth-guided convolutional block integration.
Training & Optimization	Fine-tuning model on multi-weather dataset for balanced generalization.	Adam optimizer, adaptive learning rate, data augmentation.
Validation & Evaluation	Measuring performance under rainy and low-light environments.	mAP, precision-recall metrics, confusion matrix analysis.

The fusion of RGB and depth information was implemented by concatenating feature maps at mid-level convolution layers. The model was trained using a diverse dataset of pothole images captured during monsoon and nighttime conditions to improve its resilience and adaptability.

3. Results

The performance of the proposed Depth-Guided YOLO model was evaluated against baseline YOLOv8 and other CNN-based detectors.

Table 2. Comparison of Detection Results

Model	Accuracy (%)	Precision (%)	Recall (%)	FPS (Frame Rate)
Standard YOLOv8	84.5	86.1	80.2	50
Depth-Guided YOLO (Proposed)	**92.8**	**93.5**	**90.1**	48
Faster R-CNN	78.9	82.4	75.6	32
SSD-MobileNet	80.3	83.0	77.2	55

The proposed model achieved a **92.8% detection accuracy**, outperforming traditional YOLO and CNN-based models, while maintaining near real-time processing capability at **48 FPS**. It also demonstrated improved detection stability under poor visibility and heavy rainfall, significantly reducing false positives caused by puddles and reflections.

4. Discussion

The integration of depth information into YOLO-based detection significantly enhances environmental adaptability. In low-light and rainy conditions, the RGB data alone becomes unreliable due to intensity inconsistencies. Depth sensing compensates for this by providing spatial structure and elevation details, enabling more accurate surface analysis. The results reveal that combining depth cues helps the model differentiate between potholes and similar depressions filled with water, an area where traditional YOLO architectures often fail. Additionally, the use of image enhancement and normalization preprocessing improved feature clarity, aiding in better convergence during training. While the model

slightly sacrifices inference speed due to dual-stream processing, the gain in accuracy and robustness justifies the computational cost, especially for autonomous vehicles or smart road monitoring systems.

5. Conclusion

This research presents a Depth-Guided YOLO Framework capable of detecting potholes effectively in low-light and rainy conditions. By fusing RGB and depth data, the system achieves enhanced robustness and precision, overcoming challenges of traditional 2D vision systems. The framework shows promising potential for real-time deployment in smart vehicles, road maintenance automation, and municipal infrastructure management. Future work will focus on reducing latency through model compression and exploring LiDAR or stereo-based depth sources for broader adaptability.

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